

LAB 3 REPORT

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Lab Title: Loops in Python

Theory: loops in python allow us to repeat a block of code multiple times without writing it again and again. Python mainly provides **for loop** and **while loop** to perform repetitive tasks. Loops help in processing data, iterating through sequences, and automating repeated operations.

TASK 1

```
binary = input("Enter a binary number: ").strip()
decimal = int(binary, 2)
print(f"Decimal equivalent of {binary} is {decimal}")
```

Output:

```
C:\Users\Suleiman\PyCharmMiscProject\.venv
Enter a binary number: 0001
Decimal equivalent of 0001 is 1

Process finished with exit code 0
```

```
# TASK 2
sentence = input("Enter a sentence: ").lower()
vowels = 0
consonants = 0
for char in sentence:
    if char.isalpha():
        if char in 'aeiou':
            vowels += 1
        else:
            consonants += 1
print("Vowels:", vowels)
print("Consonants:", consonants)
```

Output:

```
Enter a sentence: welcome in my life , i am the temporary best friend you will ever meet
Vowels: 22
Consonants: 33

Process finished with exit code 0
```

```
# TASK 3
start = int(input("Enter starting number: "))
end = int(input("Enter ending number: "))
primes = []
total = 0
for num in range(start, end + 1):
    if num > 1:
        is_prime = True
        for i in range(2, num):
            if num % i == 0:
                is_prime = False
                break
        if is_prime:
            primes.append(num)
            total += num

print("Prime numbers:", primes)
print("Sum of primes:", total)
```

Output:

```
C:\Users\Suleiman\Pycharm\miscProject\
Enter starting number: 11
Enter ending number: 21
Prime numbers: [11, 13, 17, 19]
Sum of primes: 60

Process finished with exit code 0
```

```
# TASK 4
import random
import string
print("=== Simple Random Password Generator ===\n")
length = int(input("Enter password length (recommended 8-16): "))
characters = string.ascii_letters + string.digits + "!@#%&*^&*"
password = "".join(random.choice(characters) for _ in range(length))
print(" Your Password:", password)
```

Output:

```
=== Simple Random Password Generator ===  
  
Enter password length (recommended 8-16): 8  
Your Password: Y34VAk6a  
  
Process finished with exit code 0
```

```
# TASK 5  
for i in range(1,5):  
    for j in range(1,i+1):  
        print("x",end=" ")  
    print()  
for i in range(3,0,-1):  
    for j in range(1,i+1):  
        print("x",end=" ")  
    print()
```

Ouptut:

```
C:\Users\Suleiman\PycharmMiscProject\.venv  
*  
* *  
* * *  
* * * *  
* * *  
* *  
*  
  
Process finished with exit code 0
```

-----THE END-----

